The valuation of European Options in Uncertain Environment

Financial Mathematics (MA3116)

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Introduction

The Black–Scholes model forms the foundation of option pricing and assumes that stock prices follow the log-normal SDE:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dB_t.$$

While this model gives elegant analytical results, real markets experience rapid fluctuations, incomplete information, and sudden changes. This causes actual prices to deviate from theoretical predictions.
Classical stochastic models treat uncertainty using only probability. In practice, uncertainty also comes from imprecision or vagueness in market information, known as **fuzziness**. Such ambiguity cannot be captured by probability alone.

Fuzziness, Fuzzy Models & PDE Connection

Fuzziness in Markets: A fuzzy number $\tilde{x}$ is described by a membership function $\mu_{\tilde{x}} : \mathbb{R} \to [0, 1]$. A triangular fuzzy number is written as:

$$\tilde{x} = (x_1, x_2, x_3),$$

where x_1, x_3 are uncertainty bounds and x_2 is the most likely value.
Fuzzy Stock Price Model:

$$dR_t = rR_t dt, \quad R_0 = 1, \qquad dS_t = \mu S_t dt + \sigma S_t dB_t,$$

but S_t is treated as a fuzzy number, carrying both randomness and fuzziness.
Fuzzy expectation uses α-cuts:

$$E[\tilde{X}]_\alpha = \left[E[X_\alpha^-],\ E[X_\alpha^+]\right].$$

European Options: If S_T is fuzzy, then the payoff

$$C = \max(S_T - K, 0)$$

becomes fuzzy, producing a **range** of option prices, not a single value.
PDE Link: The Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

must be solved with fuzzy parameters, making numerical PDE methods essential.

Fuzzy Numbers and α-Cuts

Fuzzy Number Definition

A *fuzzy number* $\tilde{a}$ is characterized by membership function $\tilde{a} : \mathbb{R} \to [0, 1]$ satisfying:

- Normality: $\exists x_0 \in \mathbb{R}$ such that $\tilde{a}(x_0) = 1$
- Upper semicontinuity, fuzzy convexity, compact support

α-Cut Representation

For $\alpha \in [0, 1]$:

$$\tilde{a}_\alpha = [\tilde{a}_\alpha^-, \tilde{a}_\alpha^+] = \{x \in \mathbb{R} \mid \tilde{a}(x) \geq \alpha\}$$

Fuzzy Arithmetic via α-cuts

$$(\tilde{a} + \tilde{b})_\alpha = [\tilde{a}_\alpha^- + \tilde{b}_\alpha^-, \tilde{a}_\alpha^+ + \tilde{b}_\alpha^+]$$
$$(\lambda \tilde{a})_\alpha = [\lambda \tilde{a}_\alpha^-, \lambda \tilde{a}_\alpha^+], \quad \lambda \geq 0$$

Fuzzy Random Variables

Definition

Let $(\Omega, \mathcal{F}, P)$ be a probability space. A mapping $\tilde{X} : \Omega \to \mathcal{F}(\mathbb{R})$ is a *fuzzy random variable* if:

$$\{(\omega, x) \in \Omega \times \mathbb{R} \mid \tilde{X}(\omega)(x) \geq \alpha\} \in \mathcal{F} \otimes \mathcal{B}$$

for all $\alpha \in [0, 1]$, where $\mathcal{B}$ is the Borel σ-field.

Practical Characterization

$\tilde{X}$ is a fuzzy random variable $\iff$ the endpoint functions

$$\omega \mapsto \tilde{X}_\alpha^-(\omega) \quad \text{and} \quad \omega \mapsto \tilde{X}_\alpha^+(\omega)$$

are measurable for all $\alpha \in [0, 1]$.

Expectation of Fuzzy Random Variables

Definition of Fuzzy Expectation

For integrably bounded fuzzy random variable $\tilde{X}$, the expectation $E(\tilde{X})$ is a fuzzy number with α-cuts:

$$E(\tilde{X})_\alpha = \left[\int_\Omega \tilde{X}_\alpha^-(\omega)\, dP(\omega), \int_\Omega \tilde{X}_\alpha^+(\omega)\, dP(\omega)\right]$$

Properties

- Extends classical expectation to fuzzy setting
- Preserves linearity property
- Membership function reconstructed via:

$$E(\tilde{X})(x) = \sup_{\alpha\in[0,1]} \min\{\alpha, \mathbb{1}_{E(\tilde{X})_\alpha}(x)\}$$

Fuzzy Stock Price Process

Classical Stock Price (Risk-Neutral)

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2}\right) t + \sigma W_t\right)$$

where r = risk-free rate, σ = volatility, W_t = Brownian motion

Fuzzy Stock Price

Membership function with triangular shape:

$$\tilde{S}_t(\omega)(x) := L\left(\frac{x - S_t(\omega)}{a_t(\omega)}\right), \quad L(x) = \max\{1 - |x|, 0\}$$

Proportional Spread Assumption

$a_t(\omega) = cS_t(\omega)$ where $c \in (0, 1)$ is the **fuzzy factor**

$$\tilde{S}_{t,\alpha}(\omega) = [b^-(\alpha)S_t(\omega), b^+(\alpha)S_t(\omega)]$$

with $b^{\pm}(\alpha) = 1 \mp (1 - \alpha)c$

Fuzzy European Option Prices

Fuzzy Call Option Payoff

$$\tilde{C}_T(\omega) := e^{-rT}(\tilde{S}_T(\omega) - K) \vee 0$$

Fuzzy Black-Scholes Formula

The α-cut endpoints of fuzzy call price:

$$\tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0) = b^{\pm}(\alpha) y \Phi(z_1^{\pm}) - Ke^{-rT}\Phi(z_2^{\pm})$$

where

$$z_1^{\pm} = \frac{\log b^{\pm}(\alpha) + \log(y/K) + T(r + \sigma^2/2)}{\sigma\sqrt{T}}$$

$$z_2^{\pm} = z_1^{\pm} - \sigma\sqrt{T}$$

Fuzzy Put-Call Parity

$$\tilde{V}^{\pm}_{\tilde{P},\alpha}(y,0) = \tilde{V}^{\pm}_{\tilde{C},\alpha}(y,0) - b^{\pm}(\alpha)y + Ke^{-rT}$$

Fuzzy Expectation Framework

- Fuzzy goal function: $u : \mathbb{R}_+ \to [0, 1]$
  - Continuous and increasing
  - $u(0) = 0$ and $\lim_{x\to\infty} u(x) = 1$
- Fuzzy expectation of expected prices:

$$\tilde{E}(E(\tilde{X}_T)) := \sup_{x\in\mathbb{R}_+} \min\{E(\tilde{X}_T)(x), u(x)\}$$

- Represents buyer's/seller's subjective judgment

Rational Expected Price

Definition: A real number $x^* \in \mathbb{R}_+$ is called a *rational expected price* if:

$$\tilde{E}(\tilde{V}) = \min\{\tilde{V}(x^*), u(x^*)\}$$

where $\tilde{V} := E(\tilde{X}_T)$ is the fuzzy price of the European option.

Key Property: The rational expected price maximizes the satisfaction degree under the fuzzy goal.

Grade of Fuzzy Expectation - Call Option

For European call options, define grade $\alpha_{\tilde{C},+}$:

$$\alpha_{\tilde{C},+} := \sup\{\alpha \in [0,1] \mid u_\alpha^- \leq \tilde{V}^\alpha_{\tilde{C},+}(y,0)\}$$

Properties:

- Grade of fuzzy expectation: $\alpha_{\tilde{C},+} = \tilde{E}(\tilde{V}_{\tilde{C}}(y,0))$
- Rational expected price: $x_{\tilde{C},+} = u^-_{\alpha_{\tilde{C},+}}$
- Similarly for lower bound: $x_{\tilde{C},-} = u^-_{\alpha_{\tilde{C},-}}$

Permissible Range of Expected Prices

Buyer's permissible range for call option:

$$[x_{\tilde{C},-}, x_{\tilde{C},+}] = \{x \in \mathbb{R} \mid \tilde{V}_{\tilde{C}}(y,0)(x) \geq u(x)\}$$

Interpretation:

- Range where reliability degree of optimal price exceeds buyer's satisfaction
- Confidence interval under uncertainty (randomness + fuzziness)

Writer's permissible range for put option:

$$[x_{\tilde{P},-}, x_{\tilde{P},+}]$$

Example: Call Option

Parameters:

- Fuzzy goal: $u(x) = 1 - e^{-2x}$ for $x \geq 0$
- Exercise time: $T = 0.5$
- Volatility: $\sigma = 0.25$
- Interest rate: $r = 0.05$
- Fuzzy factor: $c = 0.05$
- Initial stock price: $y = 30$
- Strike price: $K = 35$

Example Results: Call Option

Computed grades:

$$\alpha_{\tilde{C},-} \approx 0.744808 \quad \text{and} \quad \alpha_{\tilde{C},+} \approx 0.816004$$

Permissible range of rational expected prices:

$$[x_{\tilde{C},-}, x_{\tilde{C},+}] \approx [0.68287, 0.84642]$$

Crisp case (c $\rightarrow$ 0): Expected price $= 0.7742830$ (included in interval)

This gives the buyer a confidence interval under uncertainty.

Example: Put Option

Parameters:

- Fuzzy goal: $u(x) = 1 - e^{-0.5x}$ for $x \geq 0$
- Same parameters: $T = 0.5$, $\sigma = 0.25$, $r = 0.05$, $c = 0.05$, $y = 30$, $K = 35$

Computed grades:

$$\alpha_{\tilde{P},-} \approx 0.909709 \quad \text{and} \quad \alpha_{\tilde{P},+} \approx 0.918023$$

Permissible range:

$$[x_{\tilde{P},-}, x_{\tilde{P},+}] \approx [4.80944, 5.00263]$$

Crisp case: Expected price $= 4.910130$

Self-Financing Hedging Strategy

Definition: An $\mathcal{M}_t$-predictable process $\{(\pi_t^{0,\pm}, \pi_t^{1,\pm})\}_{t\geq 0}$

- $\pi_t^{0,\pm}$: amount of bond at time t
- $\pi_t^{1,\pm}$: amount of stock at time t

Wealth process:

$$V_{t,\alpha}^{\pm} = \pi_t^{0,\pm} R_t + \pi_t^{1,\pm} \tilde{S}_{t,\alpha}^{\pm}, \quad t \geq 0$$

Self-financing condition:

$$d\pi_t^{0,\pm} R_t + d\pi_t^{1,\pm} \tilde{S}_{t,\alpha}^{\pm} = 0$$

(No inflow or outflow of capital)

Minimal Hedging Strategy

For European call option:

$$\pi_t^{0,\pm} = \Phi\left(\frac{\log b^{\pm}(\alpha) + \log(S_t/K) + (T-t)(r+\sigma^2/2)}{\sigma\sqrt{T-t}}\right)$$

$$\pi_t^{1,\pm} = -e^{-rT}K\Phi\left(\frac{\log b^{\pm}(\alpha) + \log(S_t/K) + (T-t)(r-\sigma^2/2)}{\sigma\sqrt{T-t}}\right)$$

for $t < T$, where Φ is the standard normal CDF.

Wealth process:

$$V_{t,\alpha}^{\pm} = \pi_t^{0,\pm}R_t + \pi_t^{1,\pm}\tilde{S}_{t,\alpha}^{\pm}$$

Comparison of Hedging Approaches

Feature	**Classical Hedging**	**Fuzzy Hedging**
Uncertainty type	Randomness only	Randomness + Fuzziness
Price output	Single value	Fuzzy interval
Strategy uniqueness	Unique delta	α-parameterized family
Computational complexity	Moderate	Higher
Market applicability	Stable markets	Volatile/uncertain market

Table: Comparison of hedging approaches

Conclusion

Thank You!